When Observation Becomes Evidence

Knowledge, Hypothesis Spaces, and the Dynamics of Exploration

Do you remember when a red light first came to mean *stop*?

Before that lesson, the red light was only a visible feature—a patch of color among other patches of color. At some point, however, it acquired a consequence. You learned that the light predicted how cars would move, what pedestrians would do, and whether stepping forward was safe. The same physical stimulus remained in the world, but its place in your mind changed. Red became not merely something you saw, but something you could reason from.

This transformation is so ordinary that it is easy to miss. We encounter it whenever an unfamiliar sound becomes a warning, a facial expression becomes a clue, or a dark cloud becomes a reason to expect rain. The world supplies colors, movements, sounds, and events. It does not supply their interpretation. What an observation means depends on what the observer already knows, what possibilities the observer can imagine, and what the observation would look like if one possibility rather than another were true.

This is the point at which an observation becomes evidence.

From Observation to Evidence: The Role of Internal Representation

Evidence is often spoken of as though it were a property carried by an observation: the fingerprint is evidence, the falling barometer is evidence, the child's fever is evidence. But an observation is evidence only in relation to a claim or a set of possible explanations. A fingerprint matters because it bears differently on the possibility that one person, rather than another, touched an object. A falling barometer matters because competing expectations about tomorrow's weather do not predict it equally well. These cases suggest that an observation becomes evidence when it makes some possibilities more or less plausible than they were before.

Questions and Their Hypothesis Spaces

The hypotheses available to us are not generated from nowhere. They draw on a broader knowledge base: concepts, memories, intuitions, causal expectations, and learned distinctions. From this larger store of knowledge, we construct a local *hypothesis space* for the question at hand—the set of explanations we can represent and seriously consider. Knowledge affects which hypotheses enter that space, how finely they are differentiated, which ones initially seem plausible, and what each one leads us to expect. The hypothesis space does not mechanically dictate what we will do. Its role is more basic: it structures the epistemic possibilities that are visible to us. It shapes what can appear as a question, what can count as an answer, and which observations might change the learner's mind.

The same observation can play different roles depending on what the learner can currently represent. I propose that exploration serves three different functions across these relations: constructing possibilities under novelty, investigating failed expectations under surprise, and discriminating among alternatives under structured uncertainty.

Novelty: Constructing a Space of Possibilities

Imagine arriving in a city whose public-transport system you have never used. Lights flash above the platform, tones sound, symbols appear, and people begin moving before you know why. You can observe a great deal without yet knowing which details deserve attention. A tone might announce a train, a delay, or a closing door. At first, you do not possess sufficiently differentiated hypotheses with which to decide among these possibilities.

Learning a new language presents a similar problem. When learners first hear an unfamiliar language, they may experience it as a continuous stream of sound. They do not yet know where one word ends and another begins, which variations are meaningful, or which recurring forms refer to the same thing. Their previous linguistic knowledge still matters: they may assume that sounds form units, that repeated expressions have related meanings, and that speakers' reactions provide clues. But these general expectations do not yet amount to a well-structured set of hypotheses about the new language.

Novelty can attract attention, but attention solves only the first problem: what to process, not how to interpret it. The initial orienting response may be brief and largely automatic. What matters for learning is what happens after that moment. The learner may start with basic questions: *What is that?* or *What does this symbol mean?* If repeated observations begin to reveal regularities, the learner may start to represent previously unnoticed distinctions. A recurring sound may become a possible word; a change in intonation may become a possible signal; a pattern of movement on the platform may become a possible rule. Observation is not yet selecting among a stable set of explanations. It is helping to construct the alternatives themselves.

This representational process can subsequently shape behavior. A language learner may listen again, imitate an expression, compare the contexts in which it occurs, or watch how other speakers respond. A traveler may wait through another announcement, follow the crowd, or compare what happens after different signals. These actions are directed, but they are not necessarily diagnostic in the strong sense of testing one explicit hypothesis against another. Their immediate purpose may instead be to find repetition, variation, boundaries, and contingencies—to discover what kinds of hypotheses are worth formulating.

Novelty therefore need not produce either purely passive reception or fully targeted, diagnostic inquiry. Between those two lies a form of structure-seeking exploration: activity guided by general expectations and emerging representations before the learner can articulate clear competing hypotheses. Before evidence can select an answer, experience may first have to help the learner form a specific question and discover what the possible answers are.

Surprise: Repairing a Model That Has Failed

Once we have learned to expect something, the world can surprise us by failing to follow the pattern. Suppose that after several trips you conclude that a particular tone means the train doors are about to close. Then one day the tone sounds, but the doors remain open. The tone itself is not new. What is surprising is that the meaning you had assigned to it no longer seems to work. Surprise therefore depends on a prior

expectation: observation can contradict us only when we were already prepared to predict what would happen.

But surprise tells us only that something may be wrong, not what the correct explanation is. Perhaps the tone has more than one function. Perhaps the doors are malfunctioning. Perhaps a staff member has interrupted the normal sequence. This is why surprise and diagnostic value are not the same. Surprise concerns the gap between what was expected and what happened. Diagnostic value concerns whether an observation helps us choose between possible explanations. A highly unusual event may fit none of the explanations we currently have. An ordinary event, by contrast, may be very useful if two explanations predict it differently.

What happens next depends on whether the learner can think of an alternative explanation. If several possibilities come to mind, exploration can become more focused. The learner might compare this tone with earlier ones, inspect the other doors, or wait to see what happens at the next station. Research with infants shows that surprise can guide exploration toward the particular part of an expectation that has failed. After seeing an object apparently defy gravity, infants were more likely to drop it; after seeing an object appear to pass through a barrier, they were more likely to bang it, an action relevant to testing its solidity (Stahl and Feigenson, 2015). These actions went beyond increased attention: they were directed toward the object and property involved in the surprising event. The findings do not tell us whether infants had already formed competing explanations, but they show that surprise can organize further investigation even when we cannot yet say that the search is fully diagnostic.

Sometimes, however, the learner knows only that the old expectation has failed. No new explanation is immediately available. In that case, the next step may simply be to watch again, repeat the event, or notice details that were previously ignored. This is not yet a test between competing hypotheses. It is a way of gathering enough information to form them. The process resembles exploration under novelty, but it begins from a different place: novelty presents a world that has not yet been organized, whereas surprise reveals a problem within an organization that already exists. Surprise may therefore lead either to testing new explanations or to constructing them. Which path it takes depends on what possibilities the learner is able to represent.

Structured Uncertainty: Discriminating Among Live Hypotheses

A third situation arises when the learner can already represent several explanations but the current evidence does not distinguish them. A houseplant may be wilting because it lacks water, receives too little light, or has damaged roots. These alternatives need not be equally plausible. Perhaps lack of water is initially the leading explanation. What matters is that more than one remains live and that each implies somewhat different consequences. Uncertainty is now structured. The question is no longer simply *What is happening?* or *Why did my prediction fail?* It is *What could I observe that would bear differently on these explanations?*

That change makes targeted exploration possible. Checking whether the soil is dry may distinguish insufficient water from some alternatives. Moving the plant to better light creates a new observation under a controlled change. Inspecting the roots may be costly, so it may be deferred until cheaper observations have been exhausted. The useful action is not necessarily the one that produces the most data. It is the one expected to change the relative credibility of the hypotheses.

Children can engage in this contrastive search. When causal evidence is confounded, preschoolers preferentially select or design interventions that isolate candidate causes (Cook et al., 2011). They also

adjust their questions to the probability distribution over possible answers: a question that efficiently divides a relatively even hypothesis space may be less useful when one explanation is already much more likely than the rest (Ruggeri et al., 2017). The structure and weighting of the alternatives matter, not merely their number.

Structured uncertainty can nevertheless become intractable. If two hypotheses make the same prediction for every available observation, no amount of repeating the same search will separate them. If the learner cannot anticipate what an intervention would reveal, its potential value may remain invisible. A large hypothesis space can invite further sampling, diffuse search, or withdrawal. Uncertainty becomes productive when the learner can represent how an attainable observation might reduce it.

Exploration Closes the Loop: From Observation to Evidence and Back

Novelty, surprise, and structured uncertainty are not three separate kinds of event, nor are they fixed stages through which every learner must pass. They describe different relations between an observation and what the learner can currently represent. An unfamiliar event may initially reveal a structure that has not yet been understood. Later, a similar event may violate an established expectation. Once several explanations become available, further observations may help distinguish among them. As the learner's hypothesis space changes, the same event can take on a different epistemic meaning.

Exploration connects these changing states. What learners already know guides what they attend to, what questions they ask, and what actions they try. These actions produce new observations, which may strengthen, weaken, or reorganize the possibilities the learner can represent. Exploration therefore closes the loop: it is guided by the learner's current representation, but it also generates the observations through which that representation can change.

Imagine moving into an unfamiliar apartment building. On the first evening, you enter a hallway and the light turns on. At first, you may not know whether the light responds to movement, to the opening of the door, or to a timer. After several similar experiences, you form a simple expectation: entering the hallway causes the light to turn on. One afternoon, however, you enter and the hallway remains dark. The event is now surprising because it conflicts with a prediction you had learned to make. You may begin to consider several explanations: perhaps the sensor is broken, perhaps it detects movement only from certain directions, or perhaps it operates only when the surrounding light is dim.

These possibilities change how you explore. You might return after sunset, approach from another direction, wave your hand near the sensor, or observe what happens when someone else enters. Suppose the light responds to movement at night but not during the day. This outcome is more than another observation: it becomes evidence because it fits the light-sensitive explanation better than the alternatives. You revise your understanding and now expect the light to respond only when the hallway is both dark and occupied. That revised expectation will guide what you notice the next time the light behaves unexpectedly.

This sequence is only one possible path. A surprising event may fail to produce an alternative explanation, leading the learner back to broader observation. A diagnostic test may also show that none of the current hypotheses is adequate, forcing the learner to construct new possibilities. Novelty, surprise, and structured uncertainty therefore describe recurring moments within learning, not a fixed sequence of stages.

When Uncertainty Becomes a Question

Knowledge is often described as the opposite of uncertainty: as one grows, the other shrinks. But knowledge also makes new uncertainty possible. Learning that weather has causes allows a dark cloud to become a prediction problem. Learning several mechanisms of disease allows a symptom to support a differential diagnosis. Knowing more may permit a rapid conclusion, but it may also reveal alternatives that a novice could not yet imagine.

The important contrast, then, is not simply between knowing and not knowing. It is between an unknown that has no articulated structure and one that has become a question. Yet even an unformed question is not an absence of thought. It may first appear as attention, hesitation, repeated observation, or the feeling that something has not yet made sense. A learner does not have to wait until enough knowledge has been acquired before inquiry can begin.

Descartes famously arrived through doubt at the claim, *I think, therefore I am*. His concern was not learning or exploration, but the reversal is useful here: doubt is not where thought disappears. To doubt is already to think, even before one knows what the answer might be. Knowledge gives uncertainty a form, but it does not grant us permission to question. It makes some questions visible, while exploration carries us toward questions we could not yet formulate.

Perhaps learning is therefore not a journey from uncertainty to certainty. It is a continuing transformation of uncertainty: from something merely noticed, to something that troubles an expectation; from a vague gap, to a question; from a question, to evidence; and from evidence, to possibilities that were previously unseen. The world supplies events, but each answer changes what the mind is able to notice in them. What new question becomes possible once we have learned to see an observation as evidence?

References

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